



Mapping the themes and intellectual structure of corporate university: co-citation and cluster analyses

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Abstract

Corporate university is a unique form of educational arrangement to accomplish an organization's goals by building the intellectual capital of its employees. In the academia, it is a relatively new concept that gained foothold during the 1990s. While it is a fertile area of research with several works unravelling its diverse aspects, only a few studies have delved into building the knowledge blocks of the concept of corporate university. Therefore, this study aims at investigating the major clusters of corporate universities obtained from the extant literature. In this study, 207 pertinent articles were retrieved from Scopus, an online electronic database. Post this, co-citation and citation analysis was performed on the 207 articles and a co-citation matrix was formulated to enlist the clusters of corporate universities. Subsequently, multi-dimensional scaling and cluster analysis techniques were used to extract the essential clusters of corporate university. Finally, five major clusters emerged: (1) corporate university as a source of competitive advantage, (2) corporate university as a layered concept, (3) reimagining corporate university through technology, (4) corporate universities: paradigms and models and (5) corporate university: performance metrics. Results of this study can act as leading insights for future studies and practitioners.

Keywords Corporate university · Multi-dimensional scaling (MDS) analysis · Cluster analysis · Co-citation analysis · Literature review

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Introduction

The existence of corporate university (CU) can be traced back to the 1950s when General Electric opened its first corporate university at Crotonville to address the issue of capability development of its employees (Alonso-Gonzalez et al. 2018). However, despite the passage of 60 years since then, it was only during the late 1990s that the concept acquired fame in the corporate world (Carton et al. 2018). Over time, organizations began to perceive the corporate university concept not only as a tool to transmit internal knowledge but also as a process to ensure a company's survival in the long term (Alonso-Gonzalez et al. 2018). In the present times, most of the established companies have their own corporate universities that dedicatedly look into their learning and developmental initiatives (Wu 2015). A dynamic business environment puts pressure on leading companies to establish corporate principles and practices to maintain the synergy. The role of technology in educational endeavors at the workplace, coupled with a shift in the educational system from being curriculum-centered to work role-centered, make CUs more pertinent. Alternative terms have been used in literature for CU, but the objective remains centered at transforming organizations into learning spaces, with the thrust on building knowledge capital (Prince and Stewart 2002; Wu 2015; Scarso 2017).

In the organizational context, knowledge is deemed to be a key input for ensuring competitive advantage in the long run. In this backdrop, corporate university emerges as a pertinent strategy that can facilitate, integrate, and disseminate knowledge within an organization to attain business goals (Vizcaya-Piñeros and Uribe-Atehortúa 2014). Furthermore, corporate universities have begun to play a dominant role in the growth of human capital in companies by encouraging the adoption of knowledge management-based strategies, which act as key differentiating factors for the attainment of competitive edge and assisting organizations to achieve their objectives (Swanson et al. 2001; Anderson 2009; Wu 2015; Alonso-Gonzalez et al. 2018). The principal driver of CU growth is the apparent relation between organization's market value and the excellence of its intellectual capital reflected in its management (Dealtry 2000a, b). However, as every sector is perturbed about shelf life of knowledge being short, constant rewiring of academic institutions is crucial (Donoghue 2008; Côté and Allahar 2011). During the past 2 decades, academicians and practitioners have appreciated the role CUs have played for human resource development across organizations, worldwide (Blass 2001; Prince and Beaver 2001; Holland and Pyman 2006; Donoghue 2008; Côté and Allahar 2011). Moreover, as CUs take different forms and roles in different companies, there is a growing emphasis on understanding their role as a driver of business results (Zhang and Boies 2018). Corporate university provides an avenue for major enterprises to incorporate development-based approaches on the lines of their strategic vision, connecting individual and team development processes to produce high potential and high performing managers across functions and levels in the company (Dealtry 2000a, b; Côté and Allahar 2011; Wu 2015). Extant review works provide insightful information on the evolution of corporate university as a concept, highlighting its relevance and facilitating critical retrospection on its sustainability (Wang et al. 2010; Lui Abel and Li 2012; Ryan et al. 2015; Alonso-Gonzalez et al. 2018).

An analysis of previous studies reveals that experienced scholars have subjectively addressed how the concept of CU was built. Literature reviews rooted in subjective analysis are generally influenced by authors' time constraint, cognitive bandwidth limit, and energy (Wang et al. 2016). Their perspective of the extant works is typically impacted by their personal orientation (Wang et al. 2016; Verma 2018). There is a likelihood that certain key

papers are omitted or misconstrued to align with the authors' research inclination. To overcome this limitation and provide key insights into ongoing studies as well as a baseline for future research in the area of CU, we have selected bibliometric techniques to ensure quantitative analysis of extant research works through citations and co-citations using multi-dimensional scaling (MDS) and cluster analysis (Pilkington and Meredith 2009; Shiau et al. 2018; Verma 2018). The bibliometric techniques act as validation tools and balance judgments made by authors (Shiau et al. 2018). In view of benefits of being quantifiable and unbiased, a literature review of extant CU research can be attained through bibliometric methods. The current investigation aims to identify high-impact academic research papers in the corporate university body of knowledge and illustrates the core knowledge structure of corporate university.

Thus, the objective of this study is to identify the areas of research that are studied when talking about corporate universities. This study attempts to trace the principal knowledge of the concept of corporate university, from 1987 to 2018. To address the above research question, journal articles on corporate university were retrieved from Scopus, which is a highly acclaimed database in the academia. Further, citation and co-citation investigation was performed on the retrieved articles of corporate university to obtain a co-citation matrix. The structure of the paper is as follows: the "Introduction" and "Literature review" sections discuss the concept of corporate university and co-citation works. The third section explicates the research methods, while the fourth section presents the results. In "Discussion" section, the results have been discussed in detail. Finally, the last section delves into implications, limitations, and conclusion of the study.

Literature review

Corporate university (CU)

In a knowledge-driven economy, the evolution and growth of corporate universities is shaped by two external factors, globalization of capital and information society. Information society propels companies to produce and transfer knowledge, whereas the globalization of capital has channelized corporate universities to align themselves with market orientation (Homan and MacPherson 2005; Parshakov and Shakina 2018). Corporate university pivots on creating structured learning spaces at the workplace and is perceived as a universal constituent of modern-day companies. It is defined as 'a dynamic and flexible entity that is constantly changing in line with management and company needs' (Dealtry 2017). Corporate universities have mushroomed in numbers; for instance, in the USA alone, their number has doubled from 1000 to 2000 between the years 1997 and 2007. Across the globe, more than 4000 companies have their own corporate universities (Rademakers 2005; BCG 2013; Rademakers 2014; People Matters 2016). This upsurge in the popularity of the concept can be attributed to a given organization's pursuit of reskilling and upskilling their workforce to meet the needs of competition, shift from the manufacturing sector to the service sector and tackle the risk of downsizing that takes place in a volatile economy (Meister 1998; Patrucco et al. 2017). The rising popularity of corporate universities may be viewed as a potential threat to the business schools that are failing to fill the gaps in their educational model. It is believed that management consultants may start addressing business shortcomings through corporate universities (Grey 2001; Zammuto 2008). As learning and knowledge

are key enablers of strategic change (Prince and Beaver 2002), learning initiatives of corporate universities have contributed substantially in supporting business goals and strategies (Holland and Pyman 2006; Qiao 2009; Antonelli et al. 2013; Wu 2015; Elazier 2016).

Due to digital disruption, corporate university has evolved from being a knowledge transfer entity to becoming an organization's learning laboratory (Côté and Allahar 2011; Masalimova et al. 2014; Ryan et al. 2015). Establishment of a corporate university is based on the principle of development, which is beyond the scope of a traditional universities. Corporate universities have consciously detangled themselves from activities like granting degrees and conducting academic research. Reasons like operational limitations and usage of traditional research to meet customized requirements of the organization can be attributed to the distancing of corporate universities from getting mainstreamed (Thompson 2000; Luna-Amaya et al. 2016). A corporate university mostly engages in activities like need assessment, designing, and delivery of training and executive development programs (Allen 2007; Lee 2010; Khan 2017). A Boston Consulting Group (BCG) report (2013) states that two reasons why boards across diverse companies support corporate universities are: tying leadership development to corporate strategy, and supporting strategy development. Holistic development cuts across all processes in an organization and should not be confused with corporate training practices (Leary and Cohen 2007; Rhéaume and Gardoni 2015).

Barley (2007) delineates the salient features of a corporate university that differentiates it from the traditional training programs. A corporate university focuses on deliberating learning strategy rather than delving into the tactics of training. They are more proactive about upcoming changes, with the bigger intention of building the organization rather than being combative to workplace challenges. Corporate university builds a rigorous assessment system that increases the accountability of stakeholders in comparison to training programs (Côté and Allahar 2011; Wu 2015). Such differentiation strategically positions corporate universities to align their work activities to an organization's objectives. Earlier, corporate universities aimed at providing operational support to training programs; however, it is only the generation-next corporate university that made a paradigmatic shift by contributing to the organizational performance. It is considered as one of the several ways of building intellectual capital at the workplace. With a profound shift in the meaning of the corporate university concept, there are three identifiable features of a corporate university (a) they are corporate-level initiatives in large, highly complex and differentiated settings, (b) they pursue continued strategic alignment and (c) they attempt to raise standards, expectations and impacts with respect to training and development (Paton et al. 2005).

Taking this perspective forward, the Boston Consulting Group report conceptualized building blocks of a successful corporate university. The model explains the strategic positioning of a corporate university. There are six building blocks of a successful strategic university—(a) ambition and objectives, which focus on funneling down of the company's goals and objectives, (b) activity scope—structuring of the corporate university to facilitate attainment of company objectives, (c) target audience and content—identifying and building content for each target body, (d) delivery model—nurturing a learning environment that assists company's needs, (e) governance and structure—handling of reporting structure, finances and facilities and (f) branding and alliances—establishing new partnerships and sedimentation of a strong brand (BCG 2013; de Sá Freire et al. 2017). These building blocks require a support system to function optimally. The BCG report categorizes six key factors of the same: (a) involvement and engagement from the top leadership, (b) alignment to company's strategy, (c) proximity to the business, (d) provision of high caliber

offerings, (e) establishment of link with employee development process, (f) assessment of value and (g) internal and external marketing (BCG 2013; de Sá Freire et al. 2017).

However, the concept of corporate universities, despite the sound rationality behind their existence, has not progressed as much as it had the potential (Johnson et al. 2003; Donoghue 2008). Key errors that the companies make while setting up CUs are—incurring substantial expenditure in physical infrastructure, initial bulk spending on information technology rather than gradual spend, much focus on developing and training the top-level management, and reduced expenditure on content enrichment (Mahmod and Minhas 2006 as cited in Rhéaume and Gardoni 2015; Homan and MacPherson 2005; Macpherson et al. 2005; Cappiello and Pedrini 2017). There is a lack of empirical works that address holistic conceptualization of functioning of CUs. Much of the existing body of research is focused on the identification of few operations of CU rather than focusing on the holistic framework of its operation (Lui Abel and Li 2012). Extant research highlights the gaps in the framework of corporate universities. For instance, the training evaluation mechanism in CU is inadequate (Qiao 2009). Also, the performance evaluation framework of CUs needs to differentiate between types of CU on the basis of its mission and degree of openness to the external environment (Cappiello and Pedrini 2017).

Co-citation analysis

The foundation of the co-citation method is the frequency count with which two research works or authors are quoted together in the same work forms (Small 1973). The philosophy behind the co-citation method is the identification and grouping of articles that can be bracketed as the same ‘research front’. A group of source documents consisting of essentials of the area or base works was created using the method of document co-citation analysis, from which the co-citation matrix was obtained. Consequently, there are two sets forming a co-citation cluster in our model (a) a set of source documents that constitutes vastly cited and co-cited referenced works and (b) a set of source documents that mentions those input documents. Classifying the input or source documents is an important stage as it needs to be as exhaustive as possible in order to completely cover the progress within the concept (Acedo et al. 2006). Post selection of source documents, we start with forming a set of citing documents. The citing documents are the ones that cite the source documents. Analysis of a bigger set of citing documents aids the researcher in gleaning broader theory movements, dissemination and connected topics.

Researchers have selectively used co-citation analysis to ascertain the core knowledge structure of a specific domain by analyzing research papers. Small (1973) proposed the concept of co-citation analysis by defining the number of occurrence of two academic papers, which are cited at the same time by other papers. Whenever there is a high citation rate of recurrence, the correlation between two documents (research papers in our case) will also be high and the documents will be more important and semantically related. The statistical techniques used to explore the network were MDS, cluster analysis and factor analysis. Factor analysis is a data reduction technique applied to condense a large number of variables into a small number of factors with some variables under each factor. Cluster analysis as a statistical technique is used for the classification of a set of variables in such a manner that variables in the same group cluster are more related (in some logic) to each other than to those in other clusters. MDS analysis assists visualizing and interpreting the collected data via a two-dimensional illustration (Shiau and Dwivedi 2012; Shiau et al. 2018). Through visualization, MDS abridges samples (e.g. multi-space samples) to

low-dimensional spaces. A co-citation analysis-based research, with all three (above-mentioned) statistical methods, was previously used in the domain areas of Management Information Systems (Culnan 1987; Andrews 2003), production and supply chain (Pilkington and Liston-Heyes 1999; Shiau et al. 2015) and e-commerce (Shiau and Dwivedi 2012).

Bibliometric analysis is a methodology for investigating the published documents within a scientific category, in a region or country (Ivanović and Ho 2017). Most of the bibliometric research in the education domain is centered on text analysis or quantitative content analysis of abstracts of articles (Assefa and Rorissa 2013; Steinhardt et al. 2017; Martí-Parreño et al. 2016; Ivanović and Ho 2017; Verger et al. 2018). Furthermore, most of the bibliometric mapping is based on text analysis, extended analysis through the word association method and association index (similarity coefficients) (Assefa and Rorissa 2013). Very few studies based on the co-citation methodology have been reported in the domain area of education (Calma and Davies 2017). The current study also makes an attempt to fill this research gap by exploring the CU domain area network, utilizing the advanced bibliometric tool including factor analysis, cluster analysis and MDS. Synthesis of literature will result in the generation of new knowledge about CU reviewed in the literature (Torraco 2005). Critical analysis of the literature will pose provocative questions (or research agenda) that give direction for future research (Torraco 2005, 2016).

Research methodology

Co-citation analysis has been used in this study to ascertain the core knowledge structure of corporate universities. To examine the knowledge structure of corporate university, the academic papers were searched for, using “corporate university” and “corporate universities” as the keyword. Scopus, the acclaimed abstract and citation database, was used to search for the relevant articles on corporate university. Scopus is a well-recognized database, which offers noteworthy research contributions from a large number of scholars and research fellows (Falagas et al. 2008). Citation analysis technique was used in this study to find the high impact research papers, whereas co-citation was used to build the co-citation matrix (Shiau et al. 2018; Verma 2018). MDS and cluster analysis techniques were subsequently utilized to investigate the co-citation matrix (Shiau et al. 2018). The examination of the co-citation matrix assisted in defining the knowledge structure of corporate university. All relevant papers published prior to October 2018 were included in the study. With the output of both the keywords coinciding in our case, 340 documents were retrieved in the primary examination. Documents such as book chapters and papers from conference proceedings were excluded from the study. Finally, 207 documents and 1545 cited papers were selected. The timeline of the retrieved documents ranged from 1987 to October 2018. Bibliometric citation analysis tool-based research studies have a significant contribution for research in social sciences (Von Wartburg et al. 2005; MacRoberts and MacRoberts 1996; Shiau et al. 2018). By concentrating on the citation association amongst research papers, the bibliometric method is able to recognize impactful and quality research contribution in a domain area (Shiau et al. 2018).

Cluster analysis

As a statistical technique for grouping, cluster analysis is used to identify similar (or differentiate heterogeneous) clusters (Punj and Stewart 1983; Hansen and Jaumard 1997; Shiau

et al. 2018). In this research, Euclidean distance was used to analyze distance, which also acts as a foundation for estimating correlation. A smaller distance between two documents specifies a sturdier correlation. Correlation between the research papers was investigated in all clusters to identify the relationships amongst clusters, along with their implications.

Hierarchical cluster analysis and multidimensional scaling

In this research, Euclidean distance, z -score standardization and Ward's cluster method were used to investigate the co-citation matrix. The outcomes from the test were used to ascertain and deliberate the central knowledge structure of all the selected academic research papers pertaining to corporate universities (Shiau et al. 2018). Visualization of data through a multi-dimensional scaling analysis is low-dimensional sample space used for positioning, classifying and analyzing correlation among data (also called a perceptual map). In the current investigation, MDS analysis was utilized to chart out the association between academic papers and identify the knowledge structure (Pratt et al. 2012; Shiau et al. 2018). Multidimensional scaling is popularly used to construct maps of authors, research papers, journals, and keywords (Small and Garfield 1985). The purpose of MDS is to ascertain items in a low dimensional space in a way such that the distance between two items demonstrates the relatedness of the items precisely. Smaller distance between two items reflects a stronger relation between them (van Eck and Waltman 2017).

Results

In the present study, 207 articles related to corporate university and 1545 references from the Scopus database from 1987 to October 2018 were collected. The search was limited to this time frame as even though the knowledge pool of CU has witnessed consistent contribution in the last one and half decade, the main attributes of CU evolved during 1987. The keywords “corporate university” and “corporate universities” were used for collating the publications. The output of these two keywords coincided in our case. Citation frequencies of the cited research papers were calculated and then the research papers were sorted in the descending order of citation frequencies to frame a co-citation matrix to extract the highly cited research papers (Shiau et al. 2018). Figure 1 presents the almanac of published

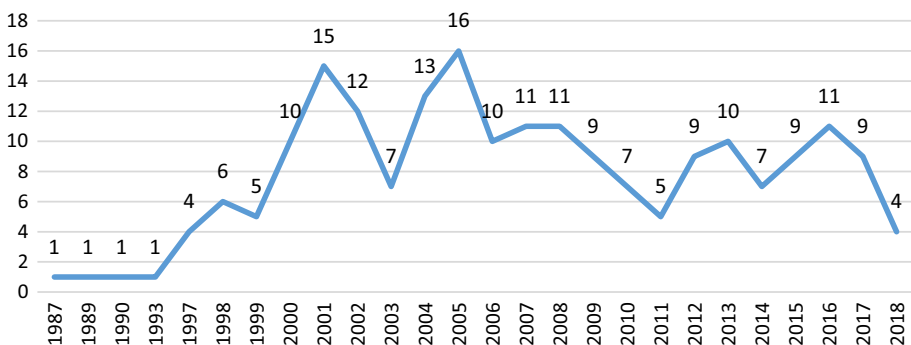


Fig. 1 Descriptive statistics of yearly academic papers in corporate university published from 1987 to October 2018

academic research papers beginning from 1987 to October 2018. It was observed that the popularity of corporate university as a research topic enhanced from the year 2000 onwards with 16 publications in 2005. Table 1 describes the spread of the top 16 Journals (with cited research papers). *Journal of Workplace Learning*, followed by the *Journal of European Industrial Training*, emerged as the leading journals that published in the field of corporate university.

Factor analysis

Retrieval of co-citation matrix

In this study, 207 research papers related to corporate university and 1545 references were retrieved from the Scopus electronic database from 1987 to October 2018. Calculation of citation frequencies of the cited articles was carried out and then the articles were sorted from highest to the lowest frequency of citation to produce a co-citation matrix in order to extract the highly cited articles. The resulting stress values from the MDS analysis were used to measure the goodness of fit of the data (Shiau et al. 2018). A low stress value (<0.2) indicates that the document data exhibit a high goodness of fit (Kruskal 1964; McCain 1990; Shiau et al. 2018). The outcomes from the study indicated that as the number of cited research papers was more or equal to 83 (stress value = 0.15142 and citation frequency = 7), the values were within the required threshold range (Kruskal 1964; McCain 1990; Shiau et al. 2018). However, as the number of cited research papers was high, the resultant MDS perceptual map was unclear. To address this issue, we performed cluster analysis to test the stress values (Shiau et al. 2018). The subsequent perceptual map met the standard of clarity, and the number of cited research papers was brought down to 76 (i.e., 36.36% of the sample), as shown in “Appendix” and Fig. 2b. Thus, 76 repeatedly

Table 1 Journals rankings with research papers contribution

Journal name	Papers
Journal of Workplace Learning	40
Journal of European Industrial Training	10
Journal of Knowledge Management	5
Industrial and Commercial Training	5
Career Development International	5
T and D	4
On the Horizon	3
Human Resource Development Quarterly	3
Cultural Studies—Critical Methodologies	3
Development and Learning in Organizations	3
Journal of Management Development	3
Higher Education	3
Critical Perspectives on Accounting	3
Human Resource Development International	3
Industry and Higher Education	3
Human Resource Management International Digest	3

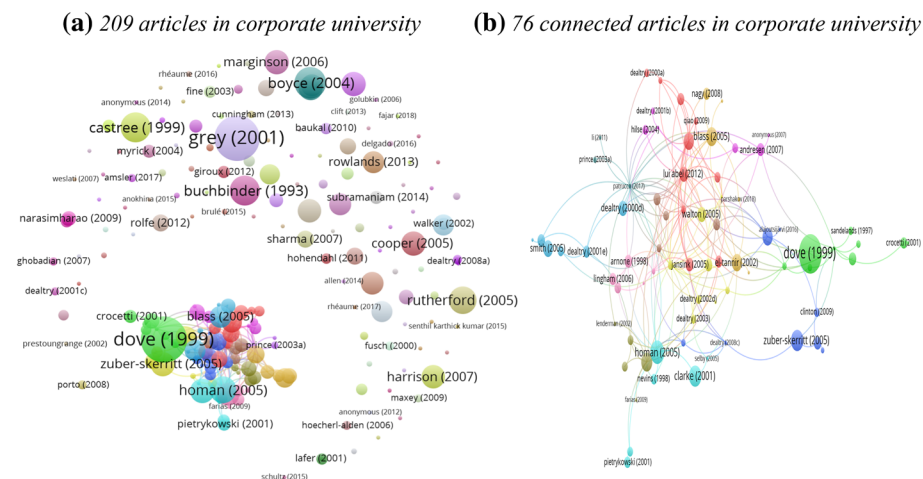


Fig. 2 **a** 209 articles in corporate university. **b** 76 connected articles in corporate university

cited documents were used to build a 76×76 co-citation matrix to follow analysis (stress value = 0.1524).

The stress value of MDS analysis was used to ascertain the possibility of choosing frequently cited research papers related to corporate university (Kruskal 1964; McCain 1990; Shiau et al. 2018). In order to describe the relationship between research papers, the co-citation matrix approach was applied. Consequently, cluster analysis was used to ascertain the key topics of corporate university. Finally, the two-dimensional (2D) perceptual map of MDS was utilized to demonstrate a research paper-research paper connection representing the fundamental knowledge of corporate university (Shiau et al. 2018) (Fig. 3).

The co-citation matrix was clustered, with the highly cited documents segregated into diverse sets, resulting in overall five clusters. Rooted in cluster analysis, MDS analysis was carried out to graphically present the relationship among the 76 often cited documents in 2D space. The relationship among documents was specifically explained through the distance between them. Figure 4 presents the results of cluster analysis and Fig. 4 depicts the multi-dimensional scaling results. Academic paper codes are shown in “Appendix”, and names of the five clusters are given below:

Cluster 1 was named corporate university as the source of competitive advantage. This cluster comprises of academic works of the following authors—Clinton et al. (2009), Andresen and Lichtenberger (2007), Ryan et al. (2015), Patrucco et al. (2017), Lenderman and Sandelands (2005), Prince and Stewart (2002), Dealtry (2001, 2002a, b) and Eccles (2004). Papers in this cluster discuss how a corporate university is different from traditional training interventions and is aligned with a company’s strategy to ensure competitive advantage.

Cluster 2 was named Corporate University as a layered concept. This cluster comprises of academic works of the following authors—Blass (2005), Homan and Macpherson (2005), Dealtry (2000a, b, 2001, 2002a, b), Walton (2005), Naggy and Robb (2008), Holland and Pyman (2006), Hilse and Nicolai (2004), Jansink et al. (2005), Prince and Stewart (2002). Papers in this cluster deliberate upon the point that corporate university has a diverse character and content. The corporate university concept derives its foundation from areas like knowledge management and organizational learning. Evolution of the

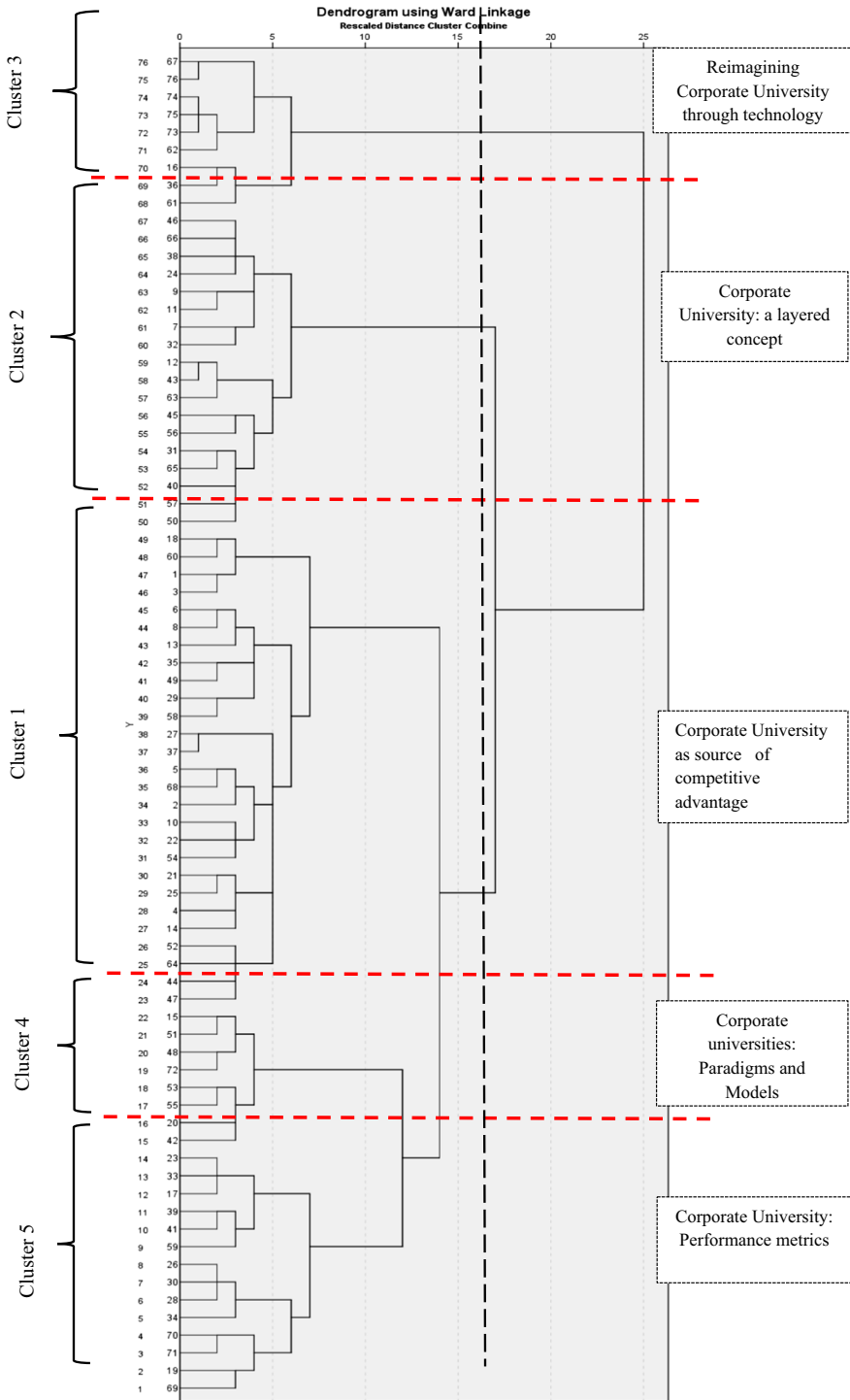


Fig. 3 Results of cluster analysis of corporate university articles

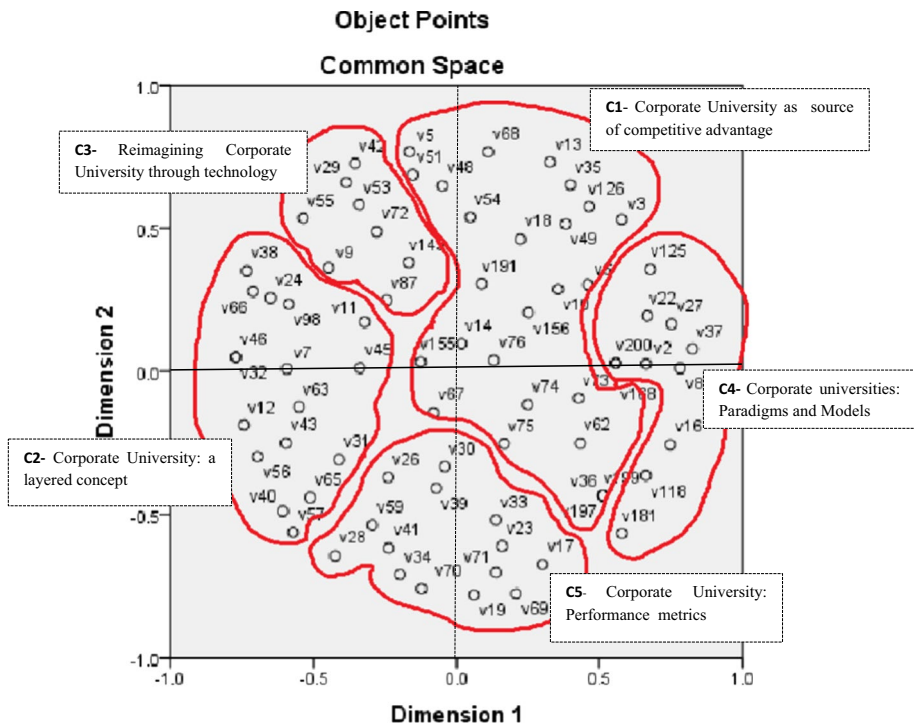


Fig. 4 Results of multi-dimensional scaling analysis of corporate university articles

concept would vary from country to country. Since the concept is heterogeneous in nature, it requires thought leadership and pro-activeness.

Cluster 3 was named Reimagining Corporate University through technology. This cluster comprises of academic works of the following authors—Clarke and Hermens (2001), Nevins (1998), Lenderman and Sandelands (2002), Roy and Elfner (2002), Farias et al. (2009), El-Tannir (2002), Selby and Russell (2005), Dealtry (2002a, b). Papers in this cluster discuss on how companies in the present times are leveraging technology in delivering the objectives of corporate universities.

Cluster 4 was named corporate universities: paradigms and models. This cluster comprises of academic works of the following authors—Prince and Allison (2003), Fulmer and Gibbs (1998), Dove (1999), Rhéaume and Gardoni (2015), Trondsen and Vickery (1997), Sandelands (1997), Ellis (1997), Matthews (1997). Papers in this cluster explore the virtual model of corporate university and the learning-on-demand model.

Cluster 5 was named corporate university: performance metrics. This cluster comprises of academic works of the following authors—Doody (2001), Lui Abel and Li (2012), Capiello and Pedrini (2017), Parshakov and Shakina (2018), Antonelli et al. (2013), Meister (1998), Ghomi and Barzinpour (2018), Arnone (1998). Papers in this cluster establish the impact of corporate universities, whether they have been a success or remain yet another cosmetic intervention by the leadership.

To visualize the relationship between the 76 articles, multidimensional scaling analysis was adopted and discussion was grounded in the output of the cluster analysis. Perceptual

map was obtained using the MDS method to illustrate the relationship between articles. According to Fig. 4, five clusters were obtained through cluster analysis and MDS analysis of reputed journal articles in the corporate university area, as outlined above. Results of this study provide penetrating insights for future studies and practitioners.

Discussion

The current investigation illustrated the yearly distribution of journals and distribution of the publications, to which the selected articles belong. Insights into the yearly publication trends direct that corporate university as an area of research began its journey from 1987 and gained foothold in the academia by 2015 with over 207 articles about the concept being published. Looking at the publication trends, it may be noted that most of the works in the area of corporate university have been published in journals. Five clusters of corporate university research emerged through cluster analysis and multidimensional scaling analysis. These clusters were (1) corporate university as a source of competitive advantage, (2) corporate university as a layered concept, (3) reimagining corporate university through technology, (4) corporate universities: paradigms and models and (5) corporate university: performance metrics. The comprehensive description of five clusters are as follows.

Cluster one: corporate university as a source of competitive advantage

Cluster 1 broadly discusses how corporate university over the years has repositioned itself as a source of competitive advantage. This cluster comprises of academic works of the following authors—Blass (2005), Homan and Macpherson (2005), Dealtry (2000a, b, 2001), Walton (2005), Naggy and Robb (2008), Holland and Pyman (2006), Hilse and Nicolai (2004), Jansink et al. (2005), Prince and Stewart (2001).

The purpose of corporate university was to fill the apparent gap between organizations and universities (Blass 2005). The promoters of such an orientation for corporate universities advocate that they need to create a sustainable competitive edge for organizations in their human resource policy. Homan and Macpherson (2005) stressed on how importance of pedagogical techniques and learner's ability towards e-learning can act as an enabler in positioning as a strategic input (Homan and Macpherson 2005). Conditioned by new demands of an increasingly dynamic and globalized market, CU over the years, has played an important role in the areas of developing and unleashing human expertise to improve individual and organizational learning and performance. During the 2000s, around 1600 corporate universities were listed in the USA, and since then, their number has increased across the globe (Dealtry 2000a, 2002b). Corporate universities of the present generation attempt the adoption of a multidisciplinary environment, enabling infrastructure, differentiating themselves from traditional universities and focusing on transferable skill sets. This helps a corporate university not only succeed but also sustain itself (Dealtry 2001; Walton 2005). Corporate universities, across nations, have gradually emerged as carriers of strategy-driven learning. They bridge the gap between the varied schools of thoughts of strategic management and corporate learning, joining general strategy with the concept of corporate universities, which has principally been an HR topic (Naggy and Robb 2008). CUs catalyze the demands for change (exogenous and endogenous) and, thus, support an organization in its quest for sustainable competitive advantage. Organization's emphasis on strategically upskilling their human capital to augment competitive advantage and the

linkage between corporate university and company's objectives are well-established (Holland and Pyman 2006; Dealtry 2000a, b).

Literature on knowledge management advocates that transfer of knowledge acts as a conduit in providing competitive advantage to a company as it facilitates seamless execution of the company's strategies (Hilse and Nicolai 2004; Clinton et al. 2009). Varied pressures from outside and inside the company often force the management not to delegate building human capital to external parties (Dealtry 2000a). Pressures include requiring just-in time solutions for different market places, external stakeholders, thinning academic rigor of management programs, established relationship between intellectual capital and shareholder goodwill value (Dealtry 2000a). It is held that alignment of type of knowledge and just media selection determines whether the corporate university would act as a competitive edge or not (Clinton et al. 2009). Though knowledge transfer and its creation are the key tenets of corporate universities, they need to reposition themselves within the organizations to maintain their relevance (Jansink et al. 2005). There is a mounting cognizance amongst managers in the major enterprises on the CU's differentiated strategic impact on their organizations. This can be considered as a result of several organizations acknowledging the benefits of knowledge and learning as central enablers of strategic change. This emerging appreciation is leading to the development of more sophisticated forms of corporate university, which are founded on the principles of knowledge management and organizational learning (Prince and Stewart 2001). The dynamic nature of companies in the modern times make them a learning space, wherein employees are often trained on work-related aspects. Corporate universities help companies achieve the same through learning management system (LMS). It is believed that an LMS should integrate training with company processes and facilitate the provision of competitive edge to the company through corporate universities (Crocetti 2001). Unfortunately, despite the increasing number of corporate universities, most of them are mere relabeling of the existing training centers. This cosmetic approach to corporate university stymies its primary objective of giving competitive advantage (Dealtry 2000a).

Cluster two: corporate university: a layered concept

Cluster 2 deliberates that corporate university is a multi-layered concept with diverse connotations ascribed to it. This cluster comprises of academic works of the following authors—Clinton et al. (2009), Andresen and Lichtenberger (2007), Ryan et al. (2015), Patrucco et al. (2017), Lenderman and Sandelands (2005), Prince and Stewart (2002), Dealtry (2001, 2002a), Eccles (2004).

There are diverse interpretations regarding what a corporate university is (Clinton et al. 2009). There is no standardized definition for corporate universities, but they may be classified as models that allow organizations to customize them to meet their training needs (Clinton et al. 2009). Researchers interpreted CU with business orientation (Andresen and Lichtenberger 2007), learning orientation (Ryan et al. 2015) and knowledge management orientation (Patrucco et al. 2017). From the business perspective, a corporate university is defined as an organization-owned learning establishment providing education that is pertinent to the business (Andresen and Lichtenberger 2007). The learning design is in alignment with the nature of a certain company's business environment and strategy and nurtures continuous learning at the individual-to-organizational level (Andresen and Lichtenberger 2007). Researchers who support CU as a learning tool believe that CU is a flexible framework within organizations to shape and implement a learning strategy that

reflects the heart of the organization's learning culture (Ryan et al. 2015). With this orientation in mind, it is vital that organizations position and market this 'learning organization' successfully within their business. Knowledge management perspective does not consider CUs as a peculiar educational or training arrangement, but regards it as a tool for managing the knowledge possessed by an organization (Patrucco et al. 2017; Ryan et al. 2015). Some authors consider them to be a knowledge management (KM) tool (Patrucco et al. 2017; Lenderman and Sandelands 2005) or even a part of the organization's knowledge infrastructure (Ryan et al. 2015). Prince and Stewart (2002) asserted that the corporate university concept branches out from existing concepts like knowledge management and organizational learning. Linking both the concepts, they proposed a corporate university framework—'corporate wheel'. Corporate wheel deliberates upon four key processes of a corporate university—(a) knowledge system and processes, (b) learning process, (c) network and partnerships and (d) people processes. It is argued that all these four processes help in attaining the objective of corporate university, which is of supporting business goals by facilitating rising expectations, sharing information, forging partnerships, creating enabling structures, and ensuring coordination.

Another level of layering takes place at the country-level application. Andresen and Lichtenberger (2007) maintain that the trajectory of corporate universities would vary from country to country. For instance, global companies in Germany took cognizance of the established relationship between aligned education and business strategy and relabeled their educational pursuits as 'corporate university'. With this shift, in the present times, most of the German companies label their traditional educational entities as corporate university. However, as one starts evaluating these entities stringently, probably only twenty companies would fit, having aligned their business strategy with education. The major difference between German and USA corporate universities is that German corporate universities are limited to their target group of management level employees, alignment to strategic and organizational change, and focus on context-specific learning strategies (Andresen and Lichtenberger 2007). Dealtry (2001, 2002a) discusses progression of the concept of corporate universities during stormy business times and explains how it requires thought leadership and proactive behavior. As corporate university managers need to influence the top management periodically, they need to possess a specific skillset. Dealtry (2002a) identifies a manager's ability to 'read and right' to classify—understand—resolve companies' key strategic and tactical issues. It is further argued that a manager should adopt real-time methodologies that help to integrate the corporate university works with overall business vision. Integration of corporate university and multi-faceted (but inter-connected) business function is imperative to keep the relevance and applicability of the CU concept (Dealtry 2002a). Engagement of all possible stakeholders and their alignment with the company's ethos is required for harnessing the multi-layered concept of CU, which has undergone constant evolution in the differing contexts (Eccles 2004).

Cluster three: reimagining corporate university through technology

Cluster 3 delves into how the adoption of technology has transformed the functioning of corporate university. This cluster comprises of academic works of the following authors—Clarke and Hermens (2001), Nevins (1998), Lenderman and Sandelands (2002), Roy and Elfner (2002), Farias et al. (2009), El-Tannir (2002), Selby and Russell (2005) and Dealtry (2002b).

Education and development are two sides of a coin, and companies are continuously striving to not only maintain but to also accelerate their learning curve (Clarke and Hermens 2001; Nevins 1998). Fulmer and Gibbs (1998) argue that in order to integrate individual development with corporate development, the amalgamation of on-the-job education, action learning and skill development is required. El-Tannir (2002) and Selby and Russell (2005) propose the adoption of information and telecommunication technology in corporate university. Blended learning, mixing online and face-to-face learning modes, has been increasingly adopted in training with the expectation that it will help solve transfer problems and will lend other educational benefits (El-Tannir 2002). The focus on corporate university should be to enable structural changes for diffusion of innovation and establish a common culture across the organization. If companies want to engage in active learning that is continuous in nature, they need to adopt future tools of training grounded in technology. Corporate universities have redefined themselves through technology optimization. The motivation to shift learning to supplementary platforms originates from an organization's desire to decrease expenses related with classroom-based training and other costs.

On similar lines, Lenderman and Sandelands (2002) advocate that companies are achieving this integration through corporate universities by transforming organizations into learning organizations. Most of the corporate universities are providing e-learning and interactive education at a global level. E-learning is an ongoing project and companies intend to migrate towards their own “knowledge communities” offer, thereby departing from the formal learning offer in order to be able to manage and operate a more “informal learning” program. Clarke and Hermens (2001) state that e-learning companies are generally setting up their own corporate universities. However, it remains to be seen that whether the CUs are facilitating the goal attainment of interactive online knowledge for communities or are they merely being used as a tool for skill training. Roy and Elfner (2002) and Farias et al. (2009) claim that though companies have directed themselves towards rigorous incorporation of IT in terms of up-gradation of instructional resources and computer labs, the research that delves into the effectiveness of such up-gradations is rather sporadic. It is imperative to ascertain the tools and techniques that are more effective than others while using technology as a medium. In order to augment its relevance, Dealtry (2002b) cites that corporate universities are focusing on offering customized structures and agile learning processes executed in a given time frame. He further proposes that customized structures require companies to focus on 3Ms: Motive, Model and Moment. “Visions must be accounted to support the formation and development of corporate universities: The Motive Vision (alignment of corporate learning to military strategies), the Model Vision (operational approach to organizational education and leadership) and the Moment Vision (the dynamics of the real-time environment)” Dealtry (2002b).

Cluster four: corporate universities: paradigms and models

Cluster 4 focuses on varied models of corporate university that have emerged over a period of time. This cluster comprises of academic works of the following authors—Prince and Allison (2003), Fulmer and Gibbs (1998), Dove (1999), Rhéaume and Gardoni (2015), Trondsen and Vickery (1997), Sandelands (1997), Ellis (1997) and Matthews (1997).

The first-generation CU were based on strategic process context (Prince and Allison 2003; Trondsen and Vickery 1997). In such universities, strategic development was a formal process, based on planned goals and actions. Such CU models were a product of interaction between organization, environment and performance. However, in the current

business dynamics, CUs are based on recurrent processes and multi-business-based generative learning across the organization. The new CUs are more entrepreneurial styled, to define and re-define emergent strategies and action-taking (Rhéaume and Gardoni 2015). The paradigms and models of corporate universities can be characterized for reinforcing and perpetuating current cultures and competitiveness as agents to manage and implement change, and as a force to drive and shape the future strategy of the organization. Success of CU models of the future could hinge more on their ability to manage and harness the complex interaction of organizational learning subsystems and less on their ability to manage training and education programs (Prince and Allison 2003; Fulmer and Gibbs 1998). Scholars suggested various models and paradigms on these lines as plug-compatible knowledge packaging model (Dove 1999; Rhéaume and Gardoni 2015), learning on demand model (Trondsen and Vickery 1997), enterprise school of management model (Sandelands 1997; Ellis 1997), Tetrahedron CU model (Matthews 1997), and corporate university wheel model (Prince and Stewart 2002; Prince and Allison 2003).

Dove (1999) proposed the concept of ‘plug-compatible knowledge packaging’, where he stressed that the concept expedites knowledge dissemination and increases the likelihood of deeper understanding for the recipient. It was further suggested that the corporate university architecture, which focuses on knowledge portfolio management and collective knowledge creation architecture, has been successfully applied in considerable cross-industry informal—group activity. Knowledge portfolio management is rooted in learning functionality and experience in organizational collaborative learning mechanisms, and shares knowledge. Trondsen and Vickery (1997) proposed the Learning-on-demand Model, which delineates that learning-on-demand reduces the time to absorb the knowledge and also cuts down on off-the job-related expenses. By harnessing modern technologies, learning-on-demand gives learners full control over the learning process and thus, the ability to coordinate the rate at which they acquire knowledge. Taking this argument further, Sandelands (1997) espoused a virtual model of corporate university, working on the principle of knowledge creation and its dissemination through electronic communication medium. Sandelands’ (1997) Enterprise School of Management Model encompassed reflective learning and integrated this with the development of learning systems. Ellis (1997), along the lines of the virtual CU model, proposed a version of the Enterprise School Management Model, which is based on the leveraging technology to deliver on the objectives of a corporate university. Matthews (1997) appreciated that a business that is going to be successful for its customers must keep technology, organization and people in a healthy balance, and expressed it as a tetrahedron model. Progress on the use of the learning organization to strengthen the tetrahedron culminates in the establishment of CU. Prince and Allison (2003) argued that the success of a corporate university is contingent on its ability to leverage the intricate interface of subsystems (Knowledge systems and processes; Network and partnership processes; People processes; and Learning processes) of organization learning, with lesser focus on handling training and educational programs.

Cluster five: corporate university: performance metrics

Cluster 5 discusses the evaluation of corporate university. This cluster comprises of academic works of the following authors—Doody (2001), Lui Abel and Li (2012), Cappiello and Pedrini (2017), Parshakov and Shakina (2018), Antonelli et al. (2013), Meister (1998, 2018) and Arnone (1998).

Corporate universities are considered to be arbitrarily expensive, and often typified as a high-risk sunken investment (Cappiello and Pedrini 2017). Nonetheless, such investments are acceptable to companies, and even produce non-operational benefits that have commercial capacity (Antonelli et al. 2013). However, it cannot be concluded that companies can set up an internal university to augment the human resource development. Hence, it is important to examine whether corporate universities create value, given that both teaching and methodology exchange can occur (Cappiello and Pedrini 2017). Also, it is vital to assess CU's ability to become an efficient investment. Strategic areas that should be focused upon for optimum effectiveness and efficiency of CU are: measurement of right alignment and execution, ensuring development of skills that support business needs, evaluation of learning and performance, optimized technology usage to support the learning function, and measuring effectiveness of partnership with academia (Doody 2001; Lui Abel and Li 2012; Cappiello and Pedrini 2017; Parshakov and Shakina 2018).

Cappiello and Pedrini (2017) considered corporate university as a hybrid entity, a combination of a higher educational institution and a business unit. Corporate universities are heterogeneous in nature and a performance framework is required to analyze the performance of CU on the basis of its mission and level of openness to the external environment (Antonelli et al. 2013; Cappiello and Pedrini 2017). Lui Abel and Li (2012) proposed that performance evaluation of corporate universities should be assessed by measuring the effect of learning efforts on organizational-level results and/or impact to the business units, and the transfer of learning to the job. In order to be successful, CUs need to establish a linkage between educational goals and company goals, develop a curriculum that stresses upon the corporate context and core competencies, expanding the scope of learning goals aligned to key customers and suppliers, and disseminating its benefits and value (Meister 1998; Ghomi and Barzinpour 2018). Parshakov and Shakina (2018) adopted economic value added and market value added as indicators to assess CUs efficiency from short-term and long-term perspectives. Such evaluation had resulted in CUs for varied impact (positive, negative or neutral) on short- or long-term performance in preceding studies (Parshakov and Shakina 2018; Arnone 1998). Doody (2001) suggested a multi-stakeholder evaluation of various aspects of CU and recommended communication of performance to stakeholders depending on the nature of their involvement. To strengthen CU assessment, independent third-party evaluations and validations can be also utilized (Doody 2001).

Implications

Researchers and practitioners will get enriched by the overall understanding of extant research in the field of corporate university. 76 noteworthy articles that have been thematically discussed here can form the base for future research regarding corporate university. From this analysis, it can be inferred that the clusters are closely connected to each other and they indicate complementarity. Cluster 1 indicates the need to assess how corporate universities repositioned themselves over the years, emerging as a source of sustainable competitive advantage. For harnessing this competitive differentiation and engagement of all stakeholders, Cluster 2 indicates company's alignment with the multi-layered concept of CU (like ethos, orientations and country-level application). In this quest of transformation, Cluster 3 underscores the significance of adopting information and telecommunication technology as a critical enabler to help accelerate this change. Analysis of the varied models of corporate university from Cluster 4 helps understand the rationale of how a

corporate university creates, delivers and captures value. It also aids in understanding the inter-linkages between different elements of any CU model. To assess whether a corporate university is creating value, Cluster 5 stresses on designing the necessary framework for university evaluation, establishing an independent mechanism to undertake the evaluation procedure and developing robust performance evaluation systems.

The business case for corporate universities is rooted in its ability to provide competitive advantage to companies. It is suggested that companies must achieve better awareness of the importance of CUs as an effective knowledge management tool (Scarso 2017). CUs have the potential to provide strategic insights and practical thinking that have influenced some of the world's leading organizations. The CU model must be sustainable to suit continuous employee learning and skill development and highlight the new approach of "active learning" that is increasingly being adopted as the future tool for training and development (El-Tannir 2002). Academicians and practitioners must understand the roles of academic and professional institutions as external forces in developing rigorous models of virtual corporate universities that meet the learning needs of individuals and organizations (Sandelands 1997). Information and communication technologies are fundamental for the development of corporate education. It may be stated that smart learning advocated by corporate universities helps the business houses maintain their uniqueness and succeed even in the face of fierce competition (Meister 1998). However, care should be taken not to overestimate corporate universities, given a few non-formalized elements herein (Chernykh and Parshikov 2016). Along with organizational readiness, CUs need to incorporate the perspectives of pedagogical as well as learner preferences into their use of e-learning, to create value in the training experience (Macpherson et al. 2005). Before establishing a corporate university, a company needs to assess the organizational readiness and learner readiness. It is noted that corporate universities fall short of achieving their objectives as they are established as a standalone intervention without assessing the readiness of different stakeholders.

As effectiveness and efficiency of corporate university is context-dependent, companies need to pay attention to the working environment of employees so that they are able to achieve the overarching company's goals. As a short-range performance resulting from investments, corporate universities might not provide long-standing benefits; hence, it is suggested that precise planning and evaluation must be carried out before establishing a corporate university (Rodriguez and Armellini 2013; Parshakov and Shakina 2017). Corporate university evaluation needs to be carried out from multi-stakeholder perspective (Doody 2001). Such an orientation is needed for two reasons, to design evaluation systems to satisfy the stakeholders, along with highlighting possible trade-offs among performances (Guerci et al. 2010). These outcomes could be valuable for crafting a connection between the human resource development (HRD) field and program evaluation. This linking can be strategic, in view of the significance of training & evaluation, which are vital for validating the value addition of HRD, and the evidence that the stakeholder participation in the process of strategic HRD alignment has a positive influence on its effectiveness (Wognum and Fond Lam 2000).

Conclusion

The primary contribution of this investigation was tracing the foundations of corporate universities, starting from 1987 to October 2018. The current research examined the core knowledge structure of the corporate university concept, its evolution and emerging trends

in the area. These results were achieved by adopting co-citation analysis, citation analysis, multi-dimensional scaling analysis and cluster analysis. The findings showed five aspects of core knowledge: (1) corporate university as a source of competitive advantage, (2) corporate university as a layered concept, (3) reimagining corporate university through technology, (4) corporate universities: paradigms and models and (5) corporate university: performance metrics. Predominantly, studies have focused on building the business case for corporate universities in view of established training and development initiatives taken by companies for decades. Our investigation highlights the role and significance of corporate universities in the central place of strategic management to integrate holistically, develop it as a source of competitive advantage and to understand the layered concept to be better equipped for reimagining corporate university through technology. Such role make-over is necessary in the dynamic business environments where an organization's competitive boundary is blurring, and its competitors are at times unpredictable. This redefined corporate university will significantly improve its assessment and development practices in terms of integrating them into the knowledge creation architecture. Corporate universities are placed as strategic business partners where they facilitate the attainment of company's goals and objectives. If the role of corporate university is organically extended, it has immense potential to catalyze an organization's sustainable competitive advantage. Corporate universities have detangled themselves from traditional training methods, as the field draws from several domains like knowledge management and learning spaces, making it a heterogeneous concept. In doing so, corporate university has repositioned itself by leveraging technology and making its offerings more interactive and cost-effective. The effectiveness of corporate universities is a query that has yielded conflicting answers, being context-dependent in most cases.

Future avenues for corporate universities research and limitations

To help enhance the academicians' interest in corporate university research and move this literature forward, the insights gleaned from this bibliometric review yield several future research directions. Given that corporate university presents its primary objective as giving competitive advantage, it is important to investigate the enablers that could provide such differentiation. As the scholars have continually revisited engagement and alignment of different stakeholders, researchers can harness this multi-layered concept of CUs to understand the underlying mechanism. Furthermore, past researchers have found different perspectives and mixed results on evaluating the performance of CU. Therefore, developing integrative frameworks for measuring the economic and market value provided by CUs and assessing CU's efficiency as per long and short-term orientations might be a fruitful avenue for future research. Also, future studies need to integrate existing and new models and paradigms to devise a comprehensive framework that can harness the complex interaction of CU learning subsystems. Lastly but most fundamentally, a significant trend of CU is associated with technologies. Therefore, researchers need to investigate the effectiveness of adoption of such technologies in transforming the functioning of corporate university.

Like any other investigation, this study also comes with certain limitations. Firstly, the bibliometric data used in the present research were collected from a single source (Scopus); hence, the omission of other research databases in the process of data collection may impact the representativeness of the data. Therefore, it is proposed that several other research databases be included to collect more comprehensive data, thus avoiding possible data bias. Secondly, only journal publications in a specific time frame (i.e., 1987–October

2018) were included; including relatively new papers with low citation frequency, books etc. may result in a better insight into the core concepts of corporate university. To the best of our knowledge, this is the first study attempting to classify highly cited and co-cited works related to Corporate Universities. Future researchers may direct their efforts to ascertain the contexts in which corporate universities are able to flourish, generating sub-themes and interdisciplinary areas of CU where we have not focused. This would act as a knowledge base for managers to address the key variables proactively, so that the concept of corporate university flourishes. Finally, the companies in the present times have vigorously adopted technology to ensure delivery of corporate university's services. This, however, raises a pertinent question that managers need to address in the future—adoption of technology to address superseding apprehension for CU's return on investment at the tradeoff of learning experience. Despite the limitations, we believe that our study provides food for thought and encouragement for researchers to further investigate the field of corporate university.

Appendix: 76 high cited corporate university articles

No.	Authors	Title	Year	Cited by
v1	Dove R.	Knowledge management, response ability, and the agile enterprise	1999	116
v2	Homan G., MacPherson A.	E-learning in the corporate university	2005	41
v3	Zuber-Skerritt O.	A model of values and actions for personal knowledge management	2005	34
v4	Clarke T., Hermens A.	Corporate developments and strategic alliances in e-learning	2001	33
v5	Macpherson A., Homan G., Wilkinson K.	The implementation and use of e-learning in the corporate university	2005	28
v6	Blass E.	The rise and rise of the corporate university	2005	27
v7	El-Tannir A.A.	The corporate university model for continuous learning, training and development	2002	24
v8	Prince C., Stewart J.	Corporate universities—an analytical framework	2002	23
v9	Blass E.	What's in a name? A comparative study of the traditional public university and the corporate university	2001	23
v10	Smith E.A.	Communities of competence: new resources in the workplace	2005	22
v11	Dealtry R.	Strategic directions in the management of the corporate university paradigm	2000	18
v12	Lui Abel A., Li J.	Exploring the corporate university phenomenon: development and implementation of a comprehensive survey	2012	17
v13	Walton J.	Would the real corporate university please stand up?	2005	17
v14	Jansink F., Kwakman K., Streumer J.	The knowledge-productive corporate university	2005	15

No.	Authors	Title	Year	Cited by
v15	Andresen M.A., Lichtenberger B.L.	The corporate university landscape in Germany	2007	13
v16	Morin L., Renaud S.	Participation in corporate university training: Its effect on individual job performance	2004	13
v17	Dealtry R.	Establishing a methodology for appraising the strategic potential of the corporate university	2000	13
v18	Meister J.C.	Ten steps to creating a corporate university	1998	13
v19	Arnone M.	Corporate universities: a viewpoint on the challenges and best practices	1998	13
v20	Trondsen E., Vickery K.	Learning on demand	1997	13
v21	Guerci M., Bartezzaghi E., Solari L.	Training evaluation in Italian corporate universities: a stakeholder-based analysis	2010	12
v22	Nagy J., Robb A.	Can universities be good corporate citizens?	2008	12
v23	Shaw S.	The corporate university: global or local phenomenon?	2005	12
v24	Dealtry R.	How to configure the corporate university for success	2001	12
v25	Lingham T., Richley B., Rezanian D.	An evaluation system for training programs: A case study using a four-phase approach	2006	11
v26	Holland P., Pyman A.	Corporate universities: a catalyst for strategic human resource development?	2006	11
v27	Dealtry R.	Configuring the corporate university—managing a portfolio of thinking schools	2001	11
v28	Pietrykowski B.	Information technology and commercialization of knowledge: corporate universities and class dynamics in an era of technological restructuring	2001	10
v29	Crocetti C.	Corporate learning—a knowledge management perspective	2001	10
v30	Stumpf S.A.	Corporate universities of the future	1998	10
v31	Bober C.F., Bartlett K.R.	The utilization of training program evaluation in corporate universities	2004	9
v32	Dealtry R.	Case research into corporate university developments	2000	9
v33	Nevins M.D.	Teaching to learn and learning to teach: notes toward building a university in a management consulting firm	1998	9
v34	Antonelli G., Cappiello G., Pedrini G.	The corporate university in the European utility industries	2013	8
v35	Hilse H., Nicolai A.T.	Strategic learning in Germany's largest companies: empirical evidence on the role of corporate universities within strategy processes	2004	8

No.	Authors	Title	Year	Cited by
v36	Eccles G.	Marketing the corporate university or enterprise academy	2004	8
v37	Campbell I., Dealtry R.	The new generation of corporate universities—co-creating sustainable enterprise and business development solutions	2003	8
v38	Roy M.H., Elfner E.	Analyzing student satisfaction with instructional technology techniques	2002	8
v39	Dealtry R.	Managing the issue of learning relevance in the formulation of corporate learning strategies	2002	8
v40	Wang G.G., Li J., Qiao X., Sun J.Y.	Understanding the corporate university phenomenon: a human capital theory perspective	2010	7
v41	Clinton M.S., Merritt K.L., Murray S.R.	Using corporate universities to facilitate knowledge transfer and achieve competitive advantage: an exploratory model based on media richness and type of knowledge to be transferred	2009	7
v42	Dealtry R.	Issues relating to learning accreditation in corporate university management	2003	7
v43	Dealtry R.	Managing the corporate university learning curve	2002	7
v44	Alagaraja M., Li J.	Utilizing institutional perspectives to investigate the emergence, rise, and (relative) decline of corporate universities	2015	6
v45	Dealtry R.	Achieving integrated performance management with the corporate university	2005	6
v46	Dealtry R.	Case research into the evolution of a corporate university development process	2000	6
v47	Sandelands E.	Developing a robust model of the virtual corporate university	1997	6
v48	Rhéaume L., Gardoni M.	The challenges facing corporate universities in dealing with open innovation	2015	5
v49	Qiao J.X.	Corporate universities in China: processes, issues and challenges	2009	5
v50	Dealtry R.	Managing the corporate university watershed	2002	5
v51	Fulmer R.M., Gibbs P.A.	Lifelong learning at the corporate university	1998	5
v52	Ellis S.	Buckman Laboratories Learning Center	1997	5
v53	Matthews P.	Aqua Universitas	1997	5
v54	Sham C.	An exploratory study of corporate universities in China	2007	4
v55	Dealtry R.	Managing the transition to the corporate university—a synthesis of client research	2001	4
v56	Dealtry R.	Interim reflections on the corporate university and SME academy business development innovation and its diffusion	2008	3

No.	Authors	Title	Year	Cited by
v57	Prince C., Allison S.	Corporate university developments in the UK SME sector	2003	3
v58	Byrd M.D.	Back to the future for higher education Medieval universities	2001	3
v59	Parshakov P., Shakina E.A.	With or without CU: a comparative study of efficiency of European and Russian corporate universities	2018	2
v60	Dealtry R.D., Howard K.H.	Moving towards optimising demand-led learning: the 2005–2007 ECUANET Leonardo Da Vinci Project	2008	2
v61	Lenderman H., Sandelands E.	Breaking the educational glass ceiling: an interview with Harry Lenderman	2005	2
v62	Lenderman H., Sandelands E.	Learning for a purpose: building a corporate university	2002	2
v63	Patrucco A.S., Pellizzoni E., Buganza T.	The design process of corporate universities: a stakeholder approach	2017	1
v64	Alajoutsijärvi K., Kettunen K.	The “Dean’s Squeeze” revisited: a contextual approach	2016	1
v65	Ryan L., Prince C., Turner P.	The changing and developing role of the corporate university post-millennium	2015	1
v66	Li J., Abel A.L.	Prioritizing + maximizing: the impact of corporate universities	2011	1
v67	Selby L., Russell D.	Curriculum design and management in the Digital Media U. Applying the corporate university concept to a business sub-sector	2005	1
v68	Dealtry R.	The real-time corporate university becomes a reality	2002	1
v69	[No author name available]	Motorola adapts training to modern business demands	2002	1
v70	Doody K.J.	Implementing a corporate-university relations program	2001	1
v71	Ghomi H., Barzinpour F.	Identifying the success factors of knowledge management tools in research projects (case study: a corporate university)	2018	1
v72	Cappiello G., Pedrini G.	The performance evaluation of corporate universities	2017	1
v73	Scarso E.	Corporate universities as knowledge management tools	2017	1
v74	Farias V., Oliveira J., Souza J.	Knowledge management integrating organizational learning in a military context: the 3M model	2009	1
v75	[No author name available]	Corporate universities: graduating with honours?	2008	1
v76	[No author name available]	Corporate universities in Germany: implementing and developing the organization’s strategy through learning	2007	1

No.	Authors	Title	Year	Cited by
v77	Dealtry R., Settle D.	Integrating programme and process performance QA	2005	1

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